STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-CARTESIAN-COFINAL-SUBCATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6277-6288 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 6290-6355, the complete author proof (64 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_sdga.json (pb=2565, stmt 6277-6288, proof 6290-6355).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSH: sources/stacks-sdga/pinned-original-source.tex, source lines 3186–3196. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GZ8: sources/stacks-sdga/pinned-original-source.tex, source lines 6105–6134. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}, \mathcal{O}, \mathcal{A}$ be as in Section 0GZ8. Let $\mathcal{C}' \subset \mathcal{C}$ be a full subcategory with the following property: for every $U \in \mathrm{Ob}(\mathcal{C})$ the category $U/\mathcal{C}'$ of arrows $U \to U'$ is cofiltered. Denote $\mathcal{O}', \mathcal{A}'$ the restrictions of $\mathcal{O}, \mathcal{A}$ to $\mathcal{C}'$. Then restrictions induces an equivalence $QC(\mathcal{A}, d) \to QC(\mathcal{A}', d)$.*

Proof. We will construct a quasi-inverse of the functor. Namely, let M' be an object of $QC(\mathcal{A}', \mathrm{d})$. We may represent M' by a good differential graded module $\mathcal{M}'$, see Lemma 0FSH. Then for every $U' \in \mathrm{Ob}(\mathcal{C}')$ the differential graded $\mathcal{A}'(U')$-module $\mathcal{M}'(U)$ is K-flat and graded flat and for every morphism $U'_1 \to U'_2$ of $\mathcal{C}'$ the map

$$\mathcal{M}'(U'_2) \otimes_{\mathcal{A}'(U'_2)} \mathcal{A}'(U'_1) \longrightarrow \mathcal{M}'(U'_1)$$

is a quasi-isomorphism (as the source represents the derived tensor product). Consider the differential graded $\mathcal{A}$-module $\mathcal{M}$ defined by the rule

$$\mathcal{M}(U) = \mathrm{colim}_{U \to U' \in U/\mathcal{C}'} \mathcal{M}'(U') \otimes_{\mathcal{A}'(U')} \mathcal{A}(U)$$

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This is a filtered colimit of complexes by our assumption in the lemma. Since M' is in $QC(\mathcal{A}', \mathrm{d})$ all the transition maps in the system are quasi-isomorphisms. Since filtered colimits are exact, we see that $\mathcal{M}(U)$ in $D(\mathcal{A}(U), \mathrm{d})$ is isomorphic to $\mathcal{M}'(U') \otimes_{\mathcal{A}'(U')} \mathcal{A}(U)$ for any morphism $U \to U'$ with $U' \in \mathrm{Ob}(\mathcal{C}')$.

We claim that $\mathcal{M}$ is in $QC(\mathcal{A}, \mathrm{d})$: namely, given $U \to V$ in $\mathcal{C}$ we choose a map $V \to V'$ with $V' \in \mathrm{Ob}(\mathcal{C}')$. By the above we see that the map $\mathcal{M}(V) \to \mathcal{M}(U)$ is identified with the map

$$\mathcal{M}'(V') \otimes_{\mathcal{A}'(V')} \mathcal{A}(V) \longrightarrow \mathcal{M}'(V') \otimes_{\mathcal{A}'(V')} \mathcal{A}(U)$$

Since $\mathcal{M}'(V')$ is K-flat as differential gradede $\mathcal{A}'(V')$-module, we conclude the claim is true.

The natural map $\mathcal{M}|_{\mathcal{C}'} \to \mathcal{M}'$ is an isomorphism in $D(\mathcal{A}', d)$ as follows immediately from the above.

Conversely, if we have an object E of $QC(\mathcal{A}, \mathrm{d})$, then we represent it by a good differential graded module $\mathcal{E}$. Setting $\mathcal{M}' = \mathcal{E}|_{\mathcal{C}'}$ (this is another good differential graded module) we see that there is a map

$$\mathcal{E} \to \mathcal{M}$$

which over U in $\mathcal{C}$ is given by the map

$$\mathcal{E}(U) \longrightarrow \mathrm{colim}_{U \to U' \in U/\mathcal{C}'} \mathcal{E}(U') \otimes_{\mathcal{A}'(U')} \mathcal{A}(U)$$

which is a quasi-isomorphism by the same reason. Thus restriction and the construction above are quasi-inverse functors as desired. □